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Abstract

This study investigates the relationship between social media usage and academic performance, as well as psychological stress, among university students. A sample of 109 students was analyzed using descriptive statistics and inferential methods including t-tests, ANOVA, correlation analysis, linear regression, and logistic regression. Although most results did not reach statistical significance, the findings provide insights into the nuanced effects of social media habits on GPA and stress levels. The analysis highlights how behavioral patterns such as frequent checking of social media during study time may serve as early indicators of academic challenges.

I. I. Introduction

Social media platforms are deeply embedded in the daily lives of university students. The pervasive use of platforms like Instagram, TikTok, and Facebook has transformed the way students communicate, consume information, and even study. As digital distractions increase, so do concerns about potential negative effects on academic performance and mental well-being. This study seeks to investigate the extent to which social media habits correlate with two critical student outcomes: GPA and self-reported stress levels. Our central aim is to understand whether patterns in usage—particularly time spent and frequency of checking social media—have any statistically measurable impact on these outcomes.

II. Methodology

A quantitative cross-sectional study was conducted involving 109 university students who completed a

structured questionnaire. Variables collected included GPA, daily hours of social media use (SM hours), frequency of checking social media while studying (check frequency), gender, stress levels, and primary reasons for using social media (e.g., for academic purposes). The responses were pre-processed by cleaning missing values, mapping categorical variables to numerical encodings, and creating dummy variables for multi-option responses. Statistical analyses were performed using Python-based packages including `scipy.stats`, `statsmodels`, and `sklearn`.

III. Results

A. Descriptive Analysis

Table 1: Descriptive Statistics of Key Variables

Variable	Mean/Percentage	Standard Deviation	Range
GPA	7.02	1.06	4.0-9.5
SM hours/day	2.7	-	-
Gender (Female)	48.6%	-	-
Gender (Male)	45.0%	-	-
Gender (Other)	6.4%	-	-

The sample consisted of 109 university students with a nearly even gender distribution. These

baseline figures provide context for the deeper statistical analyses that follow.

B. Inferential Statistical Analysis

1. Mean Comparison

Table 2: Group Comparison Results

Test	Statistic	P-value	Conclusion
t-test (Gender)	t = -0.58	0.565	No significant gender difference in GPA
ANOVA (Study Year)	F = 0.96	0.413	No significant year difference in GPA

2. Correlation Analysis

The Pearson correlation between SM hours and GPA revealed a weak negative relationship ($r = -0.18$, $p = 0.066$). Although this did not reach conventional levels of significance ($p < 0.05$), the result suggests a trend where more time on social media may be associated with slightly lower GPA outcomes.

3. Non-Parametric Testing

The Mann-Whitney U test comparing GPA between students reporting stress versus those who did not found no significant difference ($U = 362$, $p = 0.314$).

4. Linear Regression Analysis

Table 3: Linear Regression Coefficients for GPA Prediction

Predictor	Coefficient	Std. Error	t-value	P-value
Intercept	8.058	0.466	17.288	<0.001
SM hours	0.082	0.092	0.884	0.380
Check frequency	-0.397	0.179	-2.210	0.031

Predictor	Coefficient	Std. Error	t-value	P-value
Gender	-0.043	0.195	-0.220	0.826
Stress	-0.450	0.287	-1.567	0.122
Academic purpose	-0.454	1.048	-0.433	0.667

*Note: * $p < 0.05$, ** $p < 0.001$

The model explained approximately 10% of the variance in GPA ($R^2 = 0.10$, Adj. $R^2 = 0.027$).

Check frequency was the only statistically significant predictor ($p = 0.031$), supporting the hypothesis that frequent checking of social media while studying negatively impacts academic performance.

5. Logistic Regression for Stress Prediction

Table 4: Logistic Regression Results for Stress Prediction

Predictor	Coefficient	Odds Ratio	Direction
SM hours	0.314	1.369	Positive
Check frequency	-0.205	0.815	Negative
Gender	0.256	1.292	Positive
GPA	-0.259	0.772	Negative

Academic purpose	0.195	1.215	Positive
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The model achieved an accuracy of 48%, with particularly low precision (0.25) for non-stressed students, indicating challenges in stress prediction from these variables alone.

IV. Discussion

The findings of this study underscore the complexity of interpreting the impact of social

media usage on academic performance and stress. The negative coefficient for check frequency in both regression models is particularly noteworthy, suggesting that students who frequently check social media during study sessions may suffer from diminished focus and productivity.

Limitations:

- Relatively small sample size (n=109)
- Cross-sectional design limits causal inferences
- Self-reported measures may be subject to bias
- Low predictive power of the stress model (48% accuracy)

Future Directions:

- Larger, longitudinal studies
- More nuanced measures of social media use
- Inclusion of moderating variables like self-regulation

V. Conclusion

This research integrates descriptive and inferential statistical methods to examine how social media usage patterns relate to GPA and stress. While the results were largely inconclusive in terms of statistical significance, important trends were observed. Frequent checking of social media emerged as a consistent negative factor for GPA. Future research should aim to expand the sample size, incorporate longitudinal data, and include psychological or behavioral assessments to better understand causality. These findings contribute to the ongoing discussion about digital well-being and academic success in the era of constant connectivity.

Figures

- *Figure 1:* Scatterplot of SM Hours vs GPA showing the weak negative relationship ($r = -0.18$)
- *Figure 2:* Correlation Heatmap visualizing relationships between GPA, SM use, check frequency, and stress

- *Figure 3:* Boxplot of GPA by Check Frequency showing the negative association with academic performance